# Do Now · retrieval

## Lesson 067 · Divide and Conquer: Splitting the Search

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Define decomposition.

### Q2

Which input assumption does binary search require?

### Q3

How does Math.Min(a,b) behave?

### Q4

Binary search [2,4,6,8,10,12,14] for 12: show low/high and midpoint indexes.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Define and predict — Divide and conquer breaks a problem into smaller instances, solves them, and (where relevant) combines results.

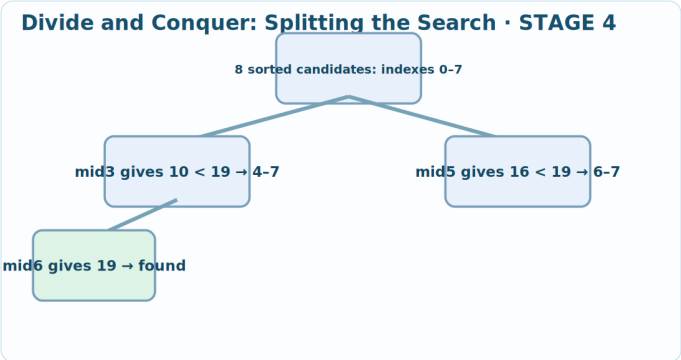
## Core idea 2

Explain — Binary search repeatedly discards half of a SORTED array. It is an example of reducing to one subproblem rather than solving both.

## Core idea 3

Evaluate — Merge sort divides into halves and combines them in sorted order. Dividing alone is not enough; correctness depends on valid preconditions and combining.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

divide and conquer — Solve smaller subproblems and combine.

precondition — Requirement for correctness.

binary search — Halving search on sorted data.

merge — Combine sorted partial results.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

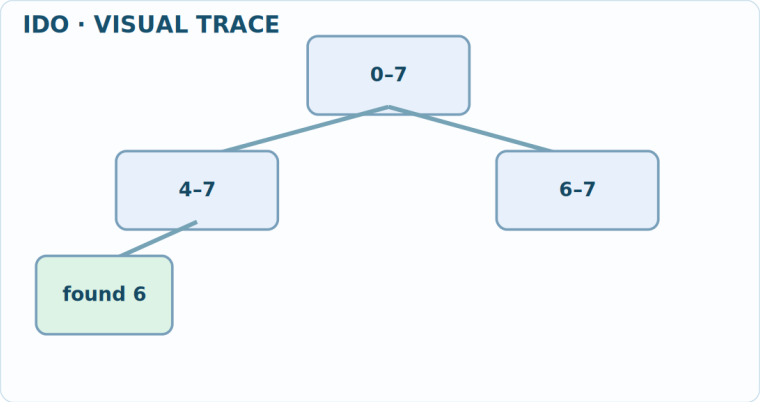
## Problem

Search sorted [2,5,8,11,14,17,20,23] for 20.

## Follow the worked method

1. Follow the reasoning — Each iteration narrows the candidate interval, and result is index 6.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Follow the reasoning
The second comparison finds the target after removing upper half.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

For [1,4,7,10,13,16,19,22] trace middle indexes when target19. [6]

## Question 2

Explain divide, solve and combine for merge sort on [7,2,9,1]. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Compare repeated halving with linear scan for sorted list of 1024 items, in terms of worst-case comparisons and preconditions. [6]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

Binary search will correctly find any target in any unsorted list.

## Use key vocabulary from memory

1. Explain divide and conquer in your own words.

2. Explain precondition in your own words.

3. Explain binary search in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.